



Society of Petroleum Engineers

SPE-227386-MS

Evaluating the Impact of Drilling Fluids on Sand Production and Mechanical Integrity of Clastic Reservoir

Osman Hamid, Saudi Aramco Dhahran Kingdom of Saudi Arabia; Gonzalo Chinea, Baker Hughes, Dhahran Saudi Arabi; Eslam Abdalrahman, King Fahad University for Petroleum and Minerals; Salem Algarni, Saudi Aramco Dhahran Kingdom of Saudi Arabia; Syed Ali, Baker Hughes, Dhahran Saudi Arabi

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227386-MS](https://doi.org/10.2118/227386-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This study investigates the effect of drilling fluids on the sanding potential of weak clastic reservoir sandstone formations. The primary objectives are to evaluate the mechanical integrity of reservoir rock after exposure to different drilling fluids and quantify the changes in sanding potential under simulated reservoir conditions.

Core samples from weak clastic reservoir were subjected to advanced thick-wall cylinder (ATWC) tests under controlled conditions of temperature (250°F), pore pressure (500 psi), and confining pressure (3000 psi). Dry samples were compared against those exposed to three different types of drilling fluids (DF1, DF2, and DF3). The physical and mechanical properties, including density, porosity, permeability, and mineralogical composition, were analyzed using XRD, XRF, and rheological data of the drilling fluids.

The study revealed a significant reduction in ATWC strength for samples exposed to drilling fluids compared to dry samples. For example, sample from study field exhibited a 17.24% reduction in ATWC when exposed to DF1. Variations in porosity and permeability were also observed, with exposed samples generally showing reduced permeability and increased density. Mineralogical analysis identified quartz as the dominant component, with minor variations in other minerals between dry and exposed conditions.

Exposure to drilling fluids reduces the sanding resistance of clastic reservoir rocks, which could influence wellbore stability and production efficiency. The degree of strength loss varies depending on the drilling fluid type and reservoir conditions, necessitating careful selection of drilling fluids to mitigate sanding risks.

This study provides a comprehensive framework to understand and quantify the impact of drilling fluids on the mechanical integrity of sandstone reservoirs. The integration of mineralogical, mechanical, and rheological data offers new insights into optimizing drilling fluid formulations to enhance wellbore stability and reduce operational risks.

Introduction

Drilling operations in deep, weak gas formations, such as clastic reservoirs, demand a comprehensive understanding of how drilling fluids interact with formation rock. These environments are especially

vulnerable to sand production due to low permeability, high in-situ stresses, and the brittle nature of the rock matrix. Sanding poses a serious threat to well integrity, manifesting as casing deformation, reduced production efficiency, equipment damage, and increased non-productive time (Zoback, 2010).

A key contributor to sanding in such reservoirs is rock weakening from drilling fluid invasion, particularly during overbalanced drilling operations. In these conditions, the hydrostatic pressure of the drilling fluid exceeds the pore pressure, enabling filtrate to invade pore spaces. This intrusion can compromise rock strength via several mechanisms: (i) swelling of clay minerals, (ii) chemical dissolution or precipitation, and (iii) weakening of grain-to-grain bonds (Zhang et al., 2011; Jaeger et al., 2007). The risk is further amplified in formations rich in reactive clays, where chemo-mechanical coupling accelerates rock failure.

The effect of different fluids on sand production has been studied by Hamid and Alfaraj 2023. In this research they provided insight into the behavior of weak sandstone formations using experimental and numerical approaches. Their investigation based on using three different pressures (Gas, oil and drilling fluid) and they identified the key parameters that control sand production in such reservoirs. Their results of the FEM modeling indicated that for the three investigation depths, an open hole wellbore filled with gas would collapse and potentially produce enough loose material to fill the wellbore.

The gas-bearing clastic formation studied here is located at depths exceeding 10,000 ft and characterized by low porosity, low permeability, and high overburden stress. Its mineralogy includes quartz, microcline, muscovite, gypsum, and minor swelling clays. The response of this reservoir to drilling fluid exposure is complex, as high temperature (up to 250°F) and pressure (up to 3000 psi) drive interactions between thermal, chemical, and mechanical processes. Even minor reductions in cohesion or rock strength from drilling fluid invasion can initiate grain mobilization, leading to progressive sand production during drawdown (Al-Ajmi & Zimmerman, 2005).

This study aims to quantify the impact of drilling fluid exposure on sanding tendency in core plugs extracted from this weak clastic reservoir. A rigorous experimental methodology was adopted, including advanced thick-walled cylinder (ATWC) tests on both dry and drilling fluid-exposed samples. Three drilling fluids (DF1, DF2, DF3) with varied chemical and electrical properties were evaluated under reservoir-representative conditions. Mineralogical (XRD) and elemental (XRF) analyses were performed to investigate postexposure transformations contributing to mechanical degradation.

Results revealed a consistent reduction in rock strength after exposure to drilling fluids, with average strength losses ranging from 15% to 23%. For instance, Sample 1 (dry) recorded an ATWC of 14,500 psi, whereas Sample 2 (after DF1 exposure) dropped to 12,000 psi—a 17.2% reduction. This weakening is attributed to increased pore pressure, reduced effective stress, and chemical weakening of grain contacts. These findings align with prior studies (Zhang et al., 2011; Al-Ajmi & Zimmerman, 2005) emphasizing the importance of drilling fluid compatibility with rock mechanics. The onset of sand production can be predicted using the following equation (Hamid and Alfaraj 2023):

$$P_P = (2(\sigma_t - \sigma_c) \tan \phi) / (3 - \sin \phi) \quad (1)$$

Where:

- P_P = the pressure required to initiate sand production
- σ_t = the tensile strength of the rock formation
- σ_c = the uniaxial compressive strength of the rock formation
- ϕ = the angle of internal friction of the rock formation

Globally, 50–60% of hydrocarbon production originates from sandstone reservoirs, and nearly 50% of these wells are at risk of sanding (Liou, 2014). The consequences range from near-wellbore damage to

equipment failure and even reservoir abandonment (Ahad, 2020). As such, managing sanding is vital for the economic viability and longevity of oil and gas assets.

To mitigate sanding, a deep understanding of the rock's mechanical behavior under realistic stress paths and chemical environments is essential. Accurate prediction requires validated mechanical models and reliable input datasets. However, the heterogeneity of weak formations poses significant modeling challenges. This study contributes to that understanding by linking compressive failure, petrophysical alteration, and fluid invasion to sand production onset providing a basis for optimized drilling fluid design and geomechanically-informed completion strategies.

Factors Influencing Sand Production

Sand production in oil and gas reservoirs is governed by a combination of geomechanical, geological, operational, and fluid interaction factors. Each contributes to the onset and severity of sanding, either by promoting rock failure or enhancing particle transport:

1. **Geomechanical Factors:** The magnitude and orientation of in-situ stresses, rock strength, and elastic moduli (e.g., Young's modulus, Poisson's ratio) are critical. High differential stress, low strength, and increased brittleness raise sanding risk (Zoback, 2010; Morita et al., 1996, and Hamid and Oluyemi 2022).
2. **Geological Factors:** Formations with weak or heterogeneous cementation, fine grain size, or high clay content are more prone to failure. Poor cementation results in lower cohesion between grains, making them easier to dislodge under stress (Fjær et al., 2008).
3. **Wellbore Geometry:** Horizontal and deviated wells induce asymmetric stress redistribution and higher near-wellbore tensile or shear stresses, which promote failure zones (Papamichos et al., 2001).
4. **Completion Design:** Suboptimal perforation density, gravel pack failure, or inadequate screen placement can trigger stress concentration, breakout zones, or ineffective sand control. The completion must be designed to maintain integrity under both static and dynamic loads (Ahad, 2020).
5. **Drilling Fluid Interaction (Damage Factor):** Drilling fluid filtrate can chemically weaken grain bonds (e.g., by feldspar dissolution, pH alteration), induce capillary stress, or trigger clay swelling. These effects reduce rock cohesion and stiffness, accelerating sanding near the wellbore (Al-Yaseri et al., 2023; Addington et al., 1991, and Hamid and Oluyemi 2022).
6. **Reservoir Pressure and Fluid Properties:** Sudden pressure drops or drawdowns beyond the critical drawdown pressure threshold can lead to tensile and compressive failure. Additionally, high-velocity flow or viscous fluids can enhance drag forces acting on detached grains (Tao et al., 2008; Morita et al., 1996).
7. **Production Strategy:** Aggressive production without controlling drawdown rates often causes early sanding. A staged depletion or managed drawdown program reduces the risk by keeping the reservoir within the safe failure envelope (Fjær et al., 2008).

Mechanism of Sand Production

Sand production is typically a two-stage process:

- **Rock Matrix Failure:** Initial weakening and degradation occur around the wellbore due to in-situ stress redistribution, drawdown-induced pressure loss, and pore pressure depletion. These lead to mechanical failure and loss of cohesion in the rock matrix (Surej Kumar, 2020).
- **Particle Mobilization and Transport:** Once failure occurs, detached grains may be transported into the wellbore by fluid forces. Hydrodynamic drag and lift act on the particles, while capillary cohesion, created by interstitial fluid surface tension, resists detachment. When reservoir pressure decreases, this cohesive resistance is overcome, allowing sand to flow with the produced fluids.

Methodology for Sand Production Prediction

A combination of experimental and numerical techniques is used to predict sand production in weak formations. One widely adopted laboratory method is the Thick-Walled Cylinder (TWC) Test.

Experimental Method: Advance Thick-Walled Cylinder (ATWC) Test

The ATWC test evaluates the mechanical behavior of rock under pressure. It provides insight into:

- Ultimate strength: The maximum internal pressure the material can withstand.
- Stress at failure: The yield strength where permanent deformation begins.
- Strain at failure: Indicates ductility and the material's ability to deform without fracturing.

If stress decreases gradually post-peak, the material exhibits toughness. In contrast, a sudden stress drop implies brittleness and low energy absorption capacity.

Critical Plastic Strain is used to quantify the onset of irreversible deformation, which is a key threshold for sand production initiation.

TWC Modeling Approaches

- Analytical Models: Solve simplified differential equations representing the stress-strain state of the cylinder. They offer quick estimations but may oversimplify complex behaviors.
- Numerical Simulations (FEM/DEM): Accurately capture the nonlinear behavior of the rock by incorporating real material heterogeneity, damage evolution, and fluid-structure interaction. These are particularly effective for calibrating predictive models under varying stress paths.

Fig. 1 Integrated geomechanical interpretation of advanced thick-walled cylinder (ATWC) test results and finite element (FE) simulations. Photographs of failed core plugs (top left and right) illustrate fracture propagation and internal hole deformation. The MohrCoulomb envelope (top center) depicts failure initiation under increasing confining pressure. Stress-strain plots (center and lower panels) represent radial and axial responses, with strain localization and asymmetry highlighting anisotropic damage evolution. Experimental data is further correlated with plastic strain accumulation and cohesion mobilization as a function of confining pressure and equivalent plastic strain. The transition from external to internal failure is calibrated against both laboratory results and FE simulations.

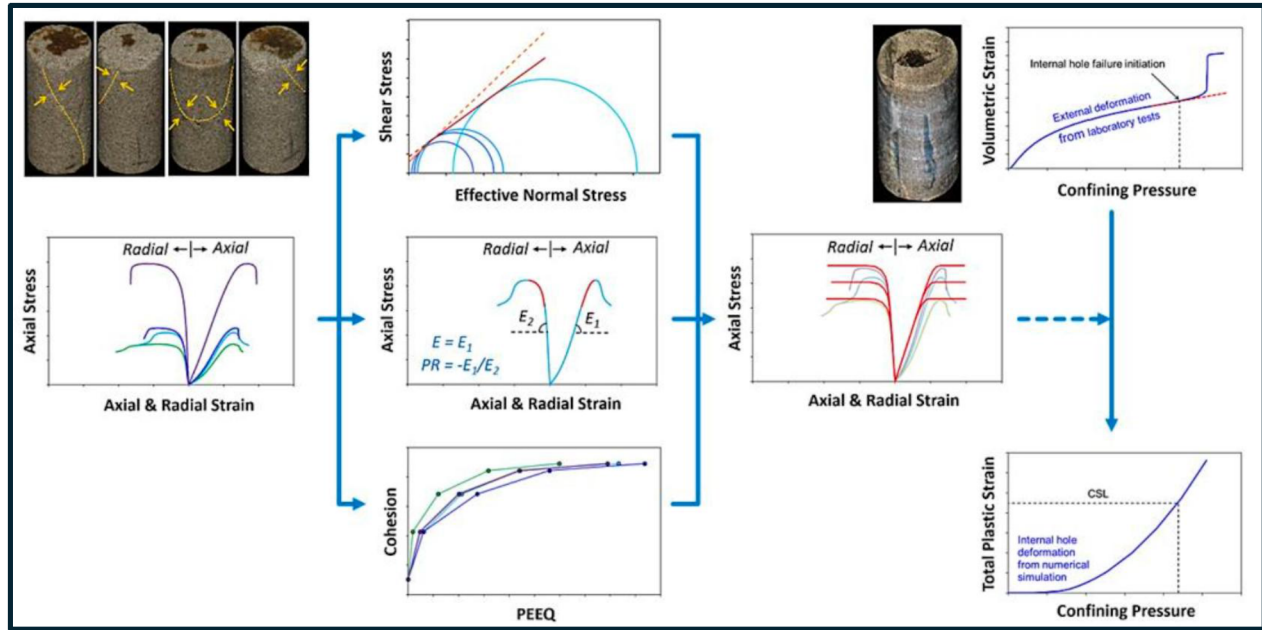


Figure 1—Integrated sand production interpretations

Numerical Formulation for Sand Prediction

Numerical modeling approaches were performed using an FEM-based poro-mechanical framework simulating:

- Rock deformation
- Pore pressure buildup
- Failure initiation and progression
- Erosion and sand transport

Equilibrium Equation

$$\sigma_{i,j,j} + f_j = 0 \quad (2)$$

Constitutive Relation

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl} \quad (3)$$

Where:

σ_{ij} is the stress tensor

ε_{kl} is strain tensor

C_{ijkl} is the stiffness tensor

Fluid Continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho V) = 0 \quad (4)$$

Where:

ρ Fluid density (kg/m^3)

V Fluid velocity vector (m/s)

$\nabla \cdot (\rho V)$ Divergence of the mass flux (rate of mass entering or leaving a control volume)

Darcy's Law (Porous Flow)

$$\mathbf{v} = -\frac{K}{\mu} \nabla P \quad (5)$$

$\mathbf{v}$: Darcy velocity (specific discharge) [m/s] (*m/s*)

K : Permeability tensor or scalar (*m²*)

μ : Dynamic viscosity of the fluid (*pa.s or cP*)

∇P : Pressure gradient driving the flow (*Pa/m*)

Sand Conservation

$$\frac{\partial(\phi \rho_s)}{\partial t} + \nabla \cdot \mathbf{q} = S_s \quad (6)$$

Where:

ϕ Porosity of the formation (volume fraction)

ρ_s Solid particle (sand) density (*kg/m³*)

$\mathbf{q}$ Sand particle flux vector [*kg/(m².s)*]

S_s Sand source term (generation or loss) [*kg/(m³.s)*]

Solid Deformation Conservation

$$\frac{\partial \varepsilon}{\partial t} + \nabla \cdot (\varepsilon \mathbf{u}) = 0 \quad (7)$$

Where:

ε Volume strain (change in volume per unit volume)

$\mathbf{u}$ Displacement vector of the solid matrix (*m*)

$\nabla \cdot (\varepsilon \mathbf{u})$ Divergence of volumetric strain transport

The assumption is the deformation of the solid matrix adheres to volume and mass conservation.

Coupled Fluid-Solid Equation

$$\mathbf{q} = -k \nabla P + \phi k \rho_g \phi (1 - \phi) \beta \mathbf{u} \quad (8)$$

Where:

$\mathbf{q}$ Total mass/momentum flux vector of the mixture

k Permeability (*m²*)

∇P Pressure gradient (*Pa /m*)

ϕ Porosity (dimensionless)

ρ_g : Body force term (gravity) (*N/m³*)

β : Solid-fluid coupling coefficient (mobility or damping term)

$\mathbf{u}$: Solid matrix displacement vector (*m*)

Sanding initiation and rate prediction

Using Mohr-Coulomb Criterion

$$\tau = \sigma_n \tan \phi + c \quad (9)$$

Sand initiation pressure (Ahad, 2020)

$$P_p = \frac{2(\sigma_t - \sigma_c) \tan \phi}{3 - \sin \phi} \quad (10)$$

Where:

σ_t : Tensile strength

σ_c : Compressive strength

ϕ : Internal friction angle

The erosion and sand grains transport can be modeled using sand momentum equation

$$\rho_p = \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} \right] = \rho_p (1 - \phi) \mathbf{g} - \nabla \cdot \boldsymbol{\sigma} \quad (11)$$

Constitutive stress for erosion is given by:

$$\sigma = -K\nabla\phi \quad (12)$$

The sand production rate equation is given by (Pham 2017)

$$Q_s = \frac{(1 - \phi)\varepsilon_v V_f}{K_f \tau_c} \quad (13)$$

Where:

Q_s : Sand production rate

ε_v : Volumetric strain

V_f : Fluid velocity

K_f : Fluid bulk modulus

τ_c : Critical shear stress

Poromechanical and Chemical Transport Formulation

The exposure setup results in fluid diffusion into the porous rock, causing:

- Changes in effective stress
- Clay swelling or dissolution
- Precipitation of secondary minerals
- Plastic deformation if yield criteria are exceeded

These processes are governed by a system of coupled partial differential equations (PDEs):

Mechanical Equilibrium with Effective Stress

$$\nabla \cdot \sigma' + \alpha \nabla P_p + \rho g = 0 \quad (14)$$

Where:

$\sigma' = \sigma - \alpha P_p \mathbf{I}$: Effective stress tensor

α : Biot coefficient

g : Gravity vector

With

$$\sigma' = C(D) : (\nabla^s \mathbf{u} - \varepsilon^p) \quad (15)$$

Where:

$C(D) = C_0(1 - D)$: Damage-modified stiffness

$\nabla^s \mathbf{u}$: Symmetric gradient of displacement

ε^p : Plastic strain

Pore pressure diffusion (Fluid flow)

$$\frac{\partial}{\partial t} \left(\alpha \nabla \cdot \mathbf{u} + \frac{P_p}{M} \right) - \nabla \cdot \left(\frac{k}{\mu} \nabla P_p \right) = Q \quad (16)$$

Where:

M : Biot modulus

k : Intrinsic permeability

μ : Fluid viscosity

Q : Source/sink term

This models fluid invasion, buildup of pore pressure, and corresponding reduction in effective normal stress σ'_n , increasing failure likelihood.

Chemical Interaction and Damage Evolution

Diffusive Transport of Reactive Species

$$\frac{\partial C}{\partial t} + \mathbf{v} \cdot \nabla C = D \nabla^2 C + R(C, T) \quad (17)$$

Where:

C : Concentration of reactive species (e.g., Na^+ , Ca^{2+} , OH^-)

D : Diffusion coefficient

$R(C, T)$: Reaction rate (e.g., for clay swelling, mineral dissolution)

This governs cation exchange, mineral instability, and fluid-rock alteration during exposure

Damage-Dependent Property Degradation

The mechanical properties degrade with damage D , evolving as:

$$E = E_0(1 - D), c = c_0(1 - D^\beta) \quad (18)$$

The damage evolution coupled to chemo-mechanical equation

$$\frac{dD}{dt} = f(\varepsilon_p, C, T) = \eta_1 \varepsilon_{eq}^p + \eta_2 C \quad (19)$$

Where:

E_0, c_0 : Initial Young's modulus and cohesion

ε_{eq}^p : Equivalent plastic strain

β : Scaling exponent

f : Empirical function capturing chemical-softening effects

C : reactive species concentration

η_1, η_2 : Weighting parameters for mechanical and chemical damage

$D \in [0, 1]$: Scalar damage variable

Damage D may initiate at the fluid-affected zone, then propagate inward with time.

Sand Detachment Threshold (Tensile and Erosion)

Once pore pressure increases and strength reduces, sand detachment can occur when:

Tensile Failure Criterion

$$\sigma'_3 < -\sigma_t \quad (20)$$

Where:

σ'_3 : Minor principal effective stress

σ_t : Tensile strength (decreases with D)

Fluid Drag Exceeds Grain Cohesion

$$F_d = \frac{1}{2} C_d \rho_f v^2 A > F_{bond} = \pi r^2 \sigma_t \quad (21)$$

if $F_d > F_{bond} = \pi r^2 \sigma_t(D)$

Where:

C_d : Drag coefficient

ρ_f : Fluid density

v : Fluid velocity

A : Projected area of a grain

r : Grain radius

$$\sigma_t(D) = \sigma_{t,0}(1 - D^\beta) \quad (22)$$

When this threshold is exceeded, fluid-induced erosion mobilizes sand, even before bulk failure.

The final numerical formulation for sand production rate $Q'_s(x, t)$

The general formulation form including AI/ML

$$Q'_s(x, t) = M^{-1} (F_{fluid} + F_{mechanical} + F_{erosion} + F_{damage} + F_{cementation} + F_{AI/ML}) \quad (23)$$

Where:

$Q'_s(x, t)$: Sand production rate (lb/day)

M^{-1} : Mechanical resistance matrix (Including rock stiffness and boundary conditions)

F_{fluid} : Fluid-induced forces

$F_{mechanical}$: Mechanical stresses and strains

$F_{erosion}$: Forces due to particle detachment and transport

F_{damage} : Damage-induced forces from material weakening

$F_{cementation}$: Reduction in resistance due to cementation degradation

$F_{AI/ML}$: AI/ML-based corrections for dynamic environmental and operational factors

The final numerical formulation for sand production rate $Q'_s(x, t)$ (lb/day)

$$Q'_s(x, t) = \rho_s \cdot A(x) \cdot \frac{(1-\phi(x,t))V_f(x,t)}{K_f \tau_c(x,t)} \cdot [\varepsilon_v(x, t) + D_{chemo}(x, t) + D_{mech}(x, t) + \Gamma_{AI/ML}(x, t) + \eta \cdot \nabla V_f(x, t) k \cdot C(x, t)] \cdot I_{yield}(x, t) \quad (24)$$

if $F_d \geq F_{bond}(x, t)$

Where:

ρ_s : Solid grain density [lb/ft³]

$A(x)$: Effective sand-transport cross sectional area [ft²]

$\phi(x, t)$: Porosity dimensionless

$V_f(x, t)$: Fluid velocity [ft/s]

K_f : Fluid bulk modulus [psi]

$\tau_c(x, t)$: Critical shear stress for sand detachment [psi]

$\varepsilon_v(x, t)$: Volumetric strain (From mechanical stress evolution)

$D_{chemo-mech}(x, t)$: Scalar damage variable (Chemical + Mechanical)

F_d : Fluid drag force (Based on velocity and pressure gradient)

F_{bond} : Bond force threshold (Cementation strength, cohesion, and grain bridging)

$\Gamma_{AI/ML}(x, t)$: Data-driven contribution (Predicted or interpolated)

$\eta \cdot \nabla V_f(x, t)$: Erosional term due to fluid acceleration or turbulence gradients

$k \cdot C(x, t)$: Compaction term: Resistance to failure from closure stress or hydrocarbon depletion

$I_{yield}(x, t)$: Yielded indicator function = 1 if the point yielded, 0 otherwise

These equations conclude:

1. Sand production occurs only when drag forces exceed bonding resistance.
2. The amount of sand produced depends on how much the rock has weakened (damage) and deformed (strain), how fast the fluid is flowing, and how loosely bonded the grains area
3. It is scaled to real-world output by including density and area, giving predictions in pounds of sand per day.

Core testing procedure

This study combines experimental testing, finite element modeling (FEM), and sand production rate estimation to assess the effect of drilling fluid exposure on sanding potential in weak sandstone formations like the weak sandstone reservoirs. Both mechanical degradation due to stress-fluid-rock interaction

and fluid-induced erosion mechanisms are considered under varying wellbore pressure and lithological conditions.

Test Procedure

Each experimental test followed a structured sequence of preparation, conditioning, and post-exposure evaluation steps designed to simulate in-situ conditions and evaluate formation damage effects on mechanical strength. Six (6) core plugs were extracted from deep tight sandstone reservoir, the samples were divided into dry baseline group and drilling fluid-exposed group, subjected to three drilling fluids with different rheology DF1, DF2, DF3. Table 1 outlines the origin and treatment conditions of core samples used for mechanical testing (e.g., ATWC), categorized into two groups, dry and drilling fluid-exposed samples, the depth matching of these samples support valid quantitative comparison.

Table 1—Sandstone core plugs information

Dry	Sample ID	1V	3V	5V
	Well	H-1	H-2	H-2
	Depth (ft)	XX455	XX100	XX105
Drilling Fluid Exposed	Sample ID	2V	4V	6V
	Well	H-1	H-2	H-2
	Depth (ft)	XX455	XX100	XX070

Sample Selection and Preparation

- Both Thick-Walled Cylinder (TWC) specimens and conventional core plugs were extracted from the same reservoir depths to ensure valid comparison between untreated (dry) and fluid-exposed conditions.
- The ends of all core plugs were precision-trimmed and surface-ground to ensure flatness and parallelism, conforming to the dimensional tolerances required for mechanical testing.

Table 2 shows core plug characterization data for both dry and drilling fluid exposed samples. Each dry sample (1V, 3 V, 5V) is paired with a corresponding fluid-exposed counterpart (2V, 4V, 6V) collected from matching reservoir depths. Physical dimensions, mass, density, porosity, and permeability are reported.

Table 2—Core plugs properties

Dry	Sample ID	1V	3V	5V
	Depth (ft)	XX455	XX100	XX105
	Diameter (mm)	37.71	37.63	37.50
	Length (mm)	78.06	78.09	78.55
	Weight (gm)	183.23	168.35	172.51
	Density (g/cc)	2.38	2.19	2.21
	Porosity	10.72	18.00	16.78
	Permeability (mD)	0.06	0.65	0.59
Drilling Fluid Exposed	Sample ID	2V	4V	6V
	Depth (ft)	XX455	XX100	XX070
	Diameter (mm)	38.02	37.95	37.61
	Length (mm)	76.08	77.49	75.31
	Weight (gm)	183.02	172.39	178.39
	Density (g/cc)	2.39	2.21	2.43
	Porosity	10.15	17.16	9.16
	Permeability (mD)	0.03	0.58	0.33

Table 3 shows elemental composition of all tested core plugs determined via XRF analysis. Key elemental oxides include Si, Al, K, and Fe, with minor components such as Mg, P, Cl, Ca, Mn, and Na. Observed variations in concentrations particularly reductions in K and S and increases in Cl and Ca indicate ion exchange, dissolution, or precipitation processes during exposure, contributing to mechanical weakening and pore blockage.

Table 3—XRF data

Minerals	Core Plug ID					
	1V	2V	3V	4V	5V	6V
Mg	0.32	0.49		0.32	0.36	0.49
Al	6.32	5.61	5.02	6.32	3.37	5.61
Si	74.98	75.68	75.00	74.98	75.39	75.68
P		0.66	0.70			0.66
S	2.46	1.98	1.56	2.46		1.98
Cl	1.59	1.33	1.60	1.59	2.85	1.33
K	9.26	7.24	5.83	9.26	9.32	7.24
Ca		1.72	1.40		3.02	1.72
Mn					0.92	
Fe	5.08	5.29	8.90	5.08	4.77	5.29

Fig. 2 shows elemental weight percentage of non-silica minerals across core plug samples 1V-6V, based on X-ray fluorescence (XRF) analysis. Quartz (Si) is excluded to highlight the distribution of less dominant, but geochemically active elements, variations in different minerals are observed across the samples.

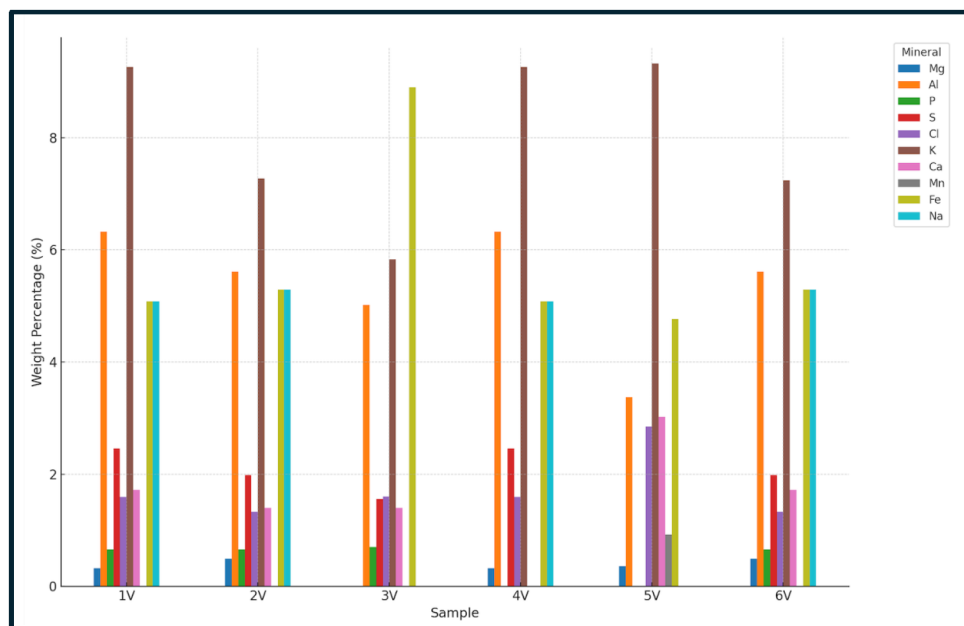


Figure 2—Mineral composition across the samples (XRF, excluding Silica)

Assembly and Confinement Setup

- The prepared core plug was mounted into the Formation Damage Simulator (FDS-800-500) system, a high-pressure cell designed for multi-physics simulation.
- Confining pressure was applied to replicate the overburden stress typically encountered at reservoir depth.

Pore Pressure and Temperature Conditioning

- Pore pressure was imposed using high-purity nitrogen gas to simulate reservoir conditions, followed by gradual heating of the system to the designated test temperature using controlled thermal input [Table 4](#).

Table 4—Tests boundary conditions

Condition	Value
Temperature (°F)	250
Pore Pressure (psi)	500
Confining Pressure (psi)	3000
Over-Balance pressure (psi)	2000

Drilling Fluid Exposure and Formation Damage Induction

- Formation damage was simulated by injecting the selected drilling fluid into the core plug in the direction of natural permeability at a constant overbalance pressure for a continuous duration of 48 hours.

- This step was designed to replicate real-world conditions where filtrate invasion, chemical interactions, and particulate retention contribute to mechanical weakening.

Table 5 summarizes the rheological and electrical properties of three drilling fluids (DF1, DF2, DF3) tested at 150°F, associated with different samples (2V, 4V, 6V). These parameters describe the flow behavior, gel strength, and electrical stability of the fluids—key properties that affect formation damage, filtrate invasion, and ultimately sand production potential.

Table 5—Drilling fluid rheology

Fluid Rheology	DF1	DF2	DF3
Sample No.	2V	4V	6V
Temperature (°F)	150	150	150
Mass weight (pcf)	89	97	101
600 RPM	63	69	79
300 RPM	41	41	47
200 RPM	31	31	36
100 RPM	21	21	25
6 RPM	9	8	10
3 RPM	7	7	9
10 sec gel	10	8	10
10 min Gel	15	15	15
Plastic Viscosity (cP)	22	28	32
Yield Point (lb/100 ft ²)	19	13	15
ES (volt)	759	699	838

Fig. 3 shows rheological properties of drilling fluids DF1, DF2, and DF3 (used on samples 2V, 4V, and 6V, respectively) tested at 150°F. Parameters include mass weight, viscometer dial readings (600-3 RPM), gel strengths (10 sec and 10 min), plastic viscosity, and yield point. DF3 exhibits the highest values across most parameters, suggesting higher cuttings suspension capability but also greater potential for formation damage. DF1 demonstrates the most formation-compatible profile with lower viscosity and stronger early gelation.

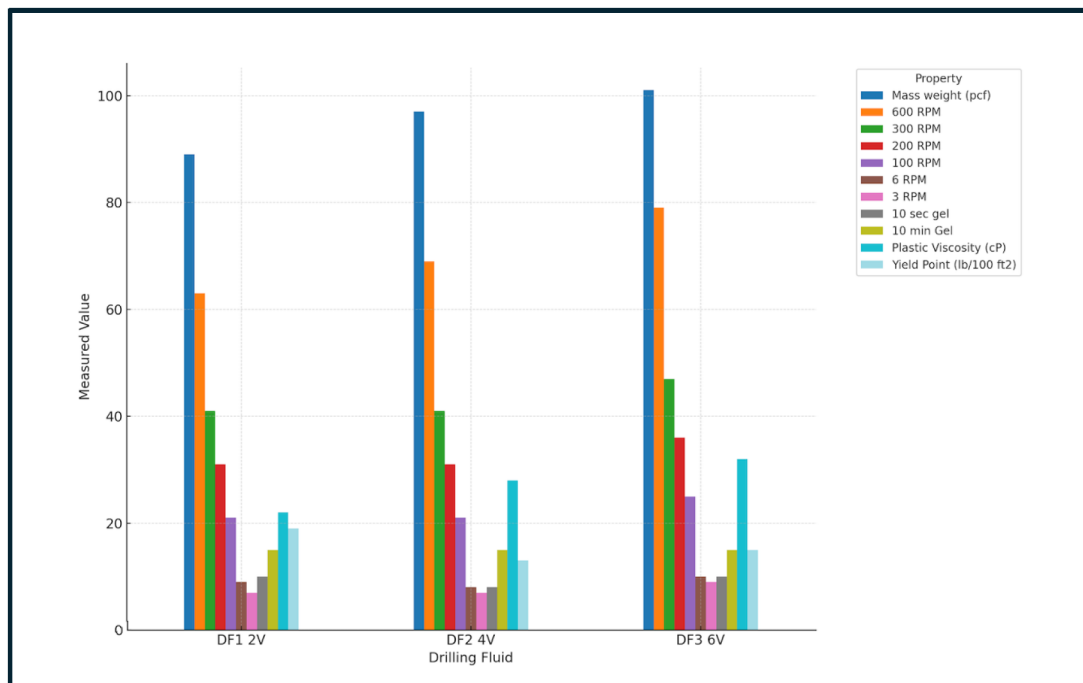


Figure 3—Drilling fluid rheology

Cool-down and Pressure Release

- Following exposure, the system was cooled down to ambient room temperature, and both pore and confining pressures were systematically released to avoid inducing artificial microfractures.

Post-Exposure Handling and Cleaning

- The exposed core plug was removed from the simulator, photographed for physical inspection, and the drilling fluid cake layer formed during exposure was carefully removed to isolate intrinsic material behavior.

TWC Sample Extraction

- The center of the exposed core plug was drilled and machined to form a TWC sample, preserving axial symmetry and capturing the zone most affected by fluid-rock interaction.

Geomechanical Testing

- The final step involved performing advanced thick-wall cylinder tests on each TWC sample to evaluate the degradation in mechanical properties (e.g., rock strength, elastic modulus) attributed to formation damage.

This procedure enabled direct comparison between pre- and post-damage mechanical responses and provided quantitative data for validating sand production and damage models.

Results

This section presents the comprehensive results of experimental evaluations conducted to determine the impact of drilling fluid exposure on the mechanical, petrophysical, mineralogical, and chemical properties of sandstone core samples extracted from the weak sandstone reservoir. The findings underscore the significant influence of drilling fluid-rock interaction on compressive failure strength, porosity and permeability

evolution, and mineral alteration, all of which directly relate to the risk of sand production during drilling and early production stages.

Mechanical Integrity: Rock Strength Degradation

Advanced thick-wall cylinder (ATWC) tests were performed on both dry and fluid-exposed samples to evaluate the impact of drilling fluids on rock strength. Results show a consistent and significant reduction in rock strength following fluid exposure, with reductions ranging from 15.9% to 22.9%. These results are summarized in Table 6.

Table 6—ATWC rock strength comparison

	Dry	DF Exposed	%Change
Sample ID	1V	2V	
ATWC (psi)	14500	12000	17.24
Sample ID	3V	4V	
ATWC (psi)	10700	9000	15.89
Sample ID	5V	6V	
ATWC (psi)	10700	8250	22.90

The mechanical weakening is attributed to several mechanisms:

Pore pressure diffusion, which reduces effective confining stress and accelerates crack initiation.

Chemical softening, particularly along grain boundaries and cement bridges, due to cation exchange, feldspar dissolution, and the precipitation of minerals like gypsum and chlorite.

Microstructural reorganization due to capillary imbibition of drilling fluids, possibly leading to localized grain detachment.

Samples exposed to DF3 consistently exhibited the greatest strength reduction, corresponding with the fluid's higher solids content, plastic viscosity (32 cP), and electrical stability (838 V), which suggests deeper and more aggressive filtrate penetration.

These changes are especially critical near the wellbore, where stress concentrations are highest and drawdown-induced shear failure may be exacerbated by reduced permeability and stiffening pore structure.

Fig. 4: presents a comparison of dry core plugs (1V, 3 V, 5V) and their corresponding DF-exposed samples (2V, 4V, 6V), tested under confined compressive loading. Rock strength values are measured in psi using the ATWC methodology. Each vertical black dashed line represents the relative reduction in rock strength following 48-hour drilling fluid exposure, expressed as a percentage loss.

- Rock strength consistently declines after DF exposure across all three samples.
- The magnitude of strength reduction ranges from ~16% to ~23%, indicating variable sensitivity of mineralogical composition and grain bonding to fluid interaction.
- Sample 6V, exposed to DF3, shows the greatest degradation, suggesting that DF3 may be chemically more aggressive or more prone to alter the microstructure.
- Sample 1V/2V shows the highest retained strength, likely due to its denser structure, less reactive clay content, or better natural cementation.

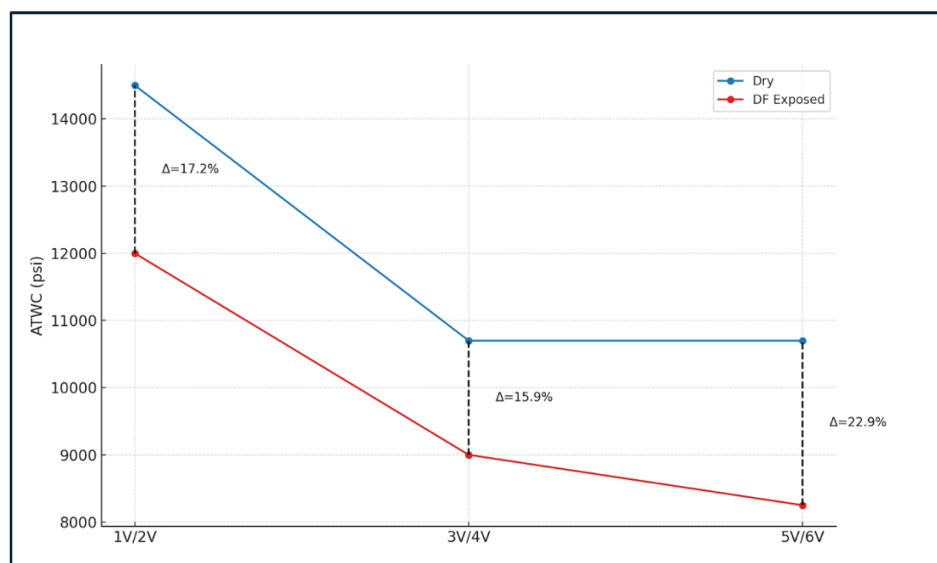


Figure 4—ATWC for dry and DF-exposed samples with % of the strength reduction

Fig. 5 Illustrate the simulation and visualizes of the relationships among porosity, permeability, advanced thick-wall cylinder (ATWC), density, and strength reduction for both dry and drilling fluid-exposed samples. A correlation matrix has been generated specifically for the exposed group, revealing how rock properties co-evolve under fluid exposure.

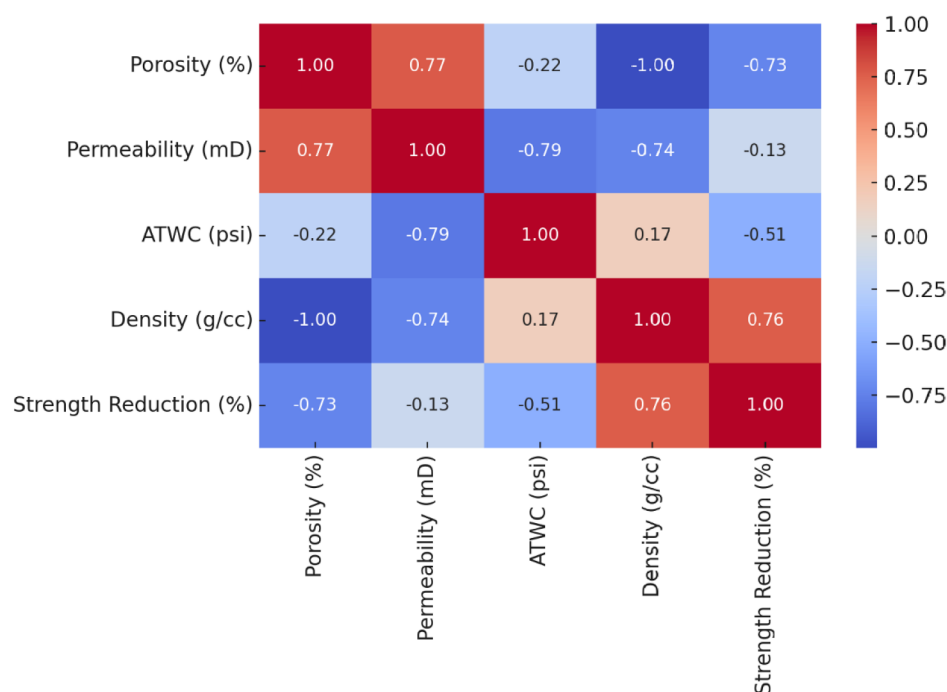


Figure 5—Correlation matrix for drilling fluid exposed samples

Key to Understanding the Matrix

- Correlation coefficients (r) range from -1 to +1:
- +1.00: perfect positive correlation

- -1.00: perfect negative correlation
- 0.00: no linear correlation
- Color scale, red: strong positive correlation, while blue: strong negative correlation

The figure shows the following correlations:

- Porosity-permeability (0.77): shows strong positive correlation. Higher porosity leads to higher permeability, as expected from pore network connectivity.
- Porosity-ATWC (-0.22) illustrate weak negative correlation. Slight decrease in rock strength with increased porosity, possibly due to loss of grain bonding.
- Porosity-density (-1.0): Shows a perfect negative correlation. As porosity increases, solid grain content (and thus density) decreases. This is physically consistent.
- Porosity-strength reduction (-0.73): Strong negative correlation. Higher porosity correlates with more mechanical damage and weakening after fluid exposure.
- Permeability-ATWC (-0.79): Strong negative correlation. Samples with higher permeability (and likely weaker cementation) show lower rock strength.
- Permeability-density (-0.74): Strong negative. Denser samples exhibit lower permeability, which is consistent with tighter grain packing.
- Permeability-strength reduction (-0.13): Very weak negative correlation. Permeability has limited influence on post-exposure strength loss.
- ATWC-density (0.17): Very weak positive correlation. Slight indication that denser samples have higher rock strength.
- ATWC-strength reduction (-0.51): Moderate negative correlation. Weaker samples (lower ATWC) suffer greater damage post-fluid exposure.
- Density-strength reduction (0.76): Strong positive correlation. Denser samples tend to lose less strength, possibly due to stronger grain framework.

Interpretation of Chemo-Mechanical Damage and Sand Production Evolution

Fig. 6 represent a realistic, time-dependent simulation of sand production risk in weak sandstone formations exposed to drilling fluids. Chemo-mechanical damage over-time (Fig. 6A)

This plot illustrates the total normalized damage index $D(t)$ resulting from combined chemical degradation and mechanical strain accumulation. The upward convex curve shows that damage evolves non-linearly, with early time dominated by rapid chemical reactions and later time reflecting compounding mechanical degradation. This progression weakens grain contacts and rock stiffness, predisposing the formation to failure. Damage approaches ~0.4 by day 7, indicating substantial structural weakening, particularly critical in weakly cemented formations.

Rock Strength Degradation (Fig. 6B)

The rock strength $\sigma_r(t)$ of the rock decreases consistently with increasing damage. This inverse relationship reflects the chemical weakening of cementing agents and bond degradation under mechanical load. A drop from ~10700 psi to ~10280 psi (a ~4% decrease) confirms progressive compressive failure potential. Even moderate reductions in rock strength can significantly increase the likelihood of sand production under drawdown conditions.

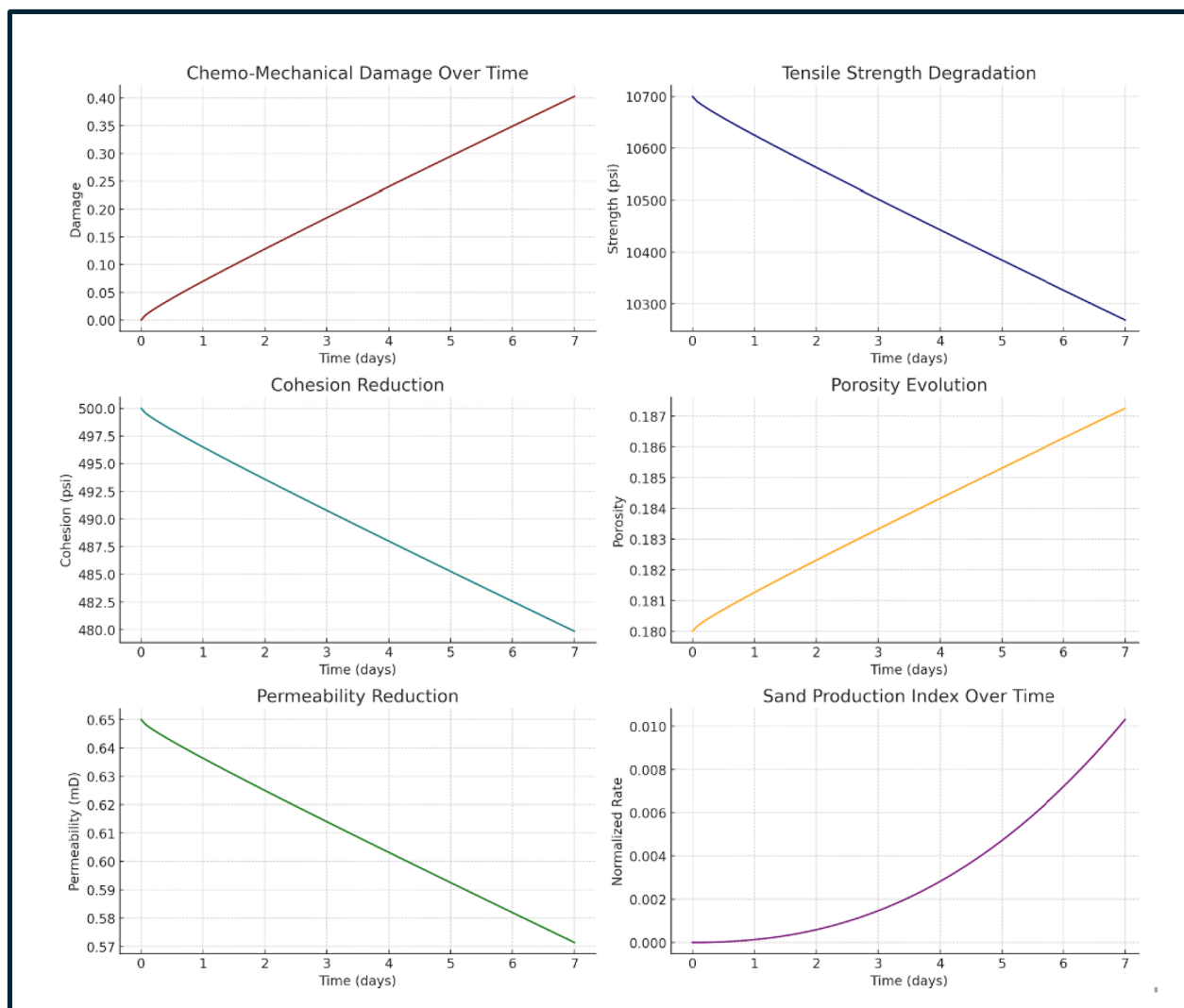


Figure 6—Time-dependent simulation of sand production risk in weak sandstone formations exposed to drilling fluids

Cohesion Reduction (Fig. 6C)

Cohesion $c(t)$, a key parameter in Mohr-Coulomb failure, declines almost linearly from 500 psi to ~480 psi. This represents the loss of intergranular bonding due to micro-fracturing and dissolution of cementing minerals (e.g., carbonates or clays) under prolonged fluid exposure. Mechanical damage and reactive drilling fluids reduce cohesion, making shear failure more likely at lower effective stresses.

Porosity Evolution (Fig. 6D)

Porosity $\phi(t)$ increases as damage progresses, rising from 0.180 to ~0.187. This reflects matrix dilatancy and grain detachment, creating new pore space. Though subtle in absolute terms (~0.007 increase), such a rise can significantly enhance fluid pathways and destabilize the formation structure. Porosity increase feeds back into permeability and alters stress redistribution in the formation.

Permeability Reduction (Fig. 6E)

Despite increasing porosity, permeability $k(t)$ declines steadily. This counterintuitive result arises from fine migration, clay swelling, and pore throat constriction. The permeability drops from 0.65 mD to ~0.57 mD (~12% reduction), impairing fluid flow and possibly promoting near-wellbore pressure buildup. Reactive fluids may increase porosity while clogging flow paths, reducing effective permeability.

Sand Production Index Over Time (Fig. 6F)

This normalized index aggregates the preceding variables to estimate sand production risk. It exhibits exponential-like growth, increasing from near zero to >0.01 by day 7. The shape suggests that a critical threshold is crossed around Day 4 -5, after which sand mobilization accelerates. Conclusion: Sand production is not linear—it remains negligible initially, then rises sharply due to cumulative damage and failure processes.

Fig. 7 is the updated plot of sand production rate $\dot{Q}_s(x, t)$ over 7 days for spatial locations from $x = 0.1$ to 1.0. Each curve represents a different position along the spatial axis.

- Most curves show a consistent increase in sand production over time, driven by cumulative damage and strain.
- A few locations (e.g., around $x = 0.8$) show near-zero sand production, indicating that the drag force was insufficient to exceed the bond strength in those zones.
- This emphasizes the spatial variability of sanding risk, crucial for FEM-based wellbore simulations and completion design.

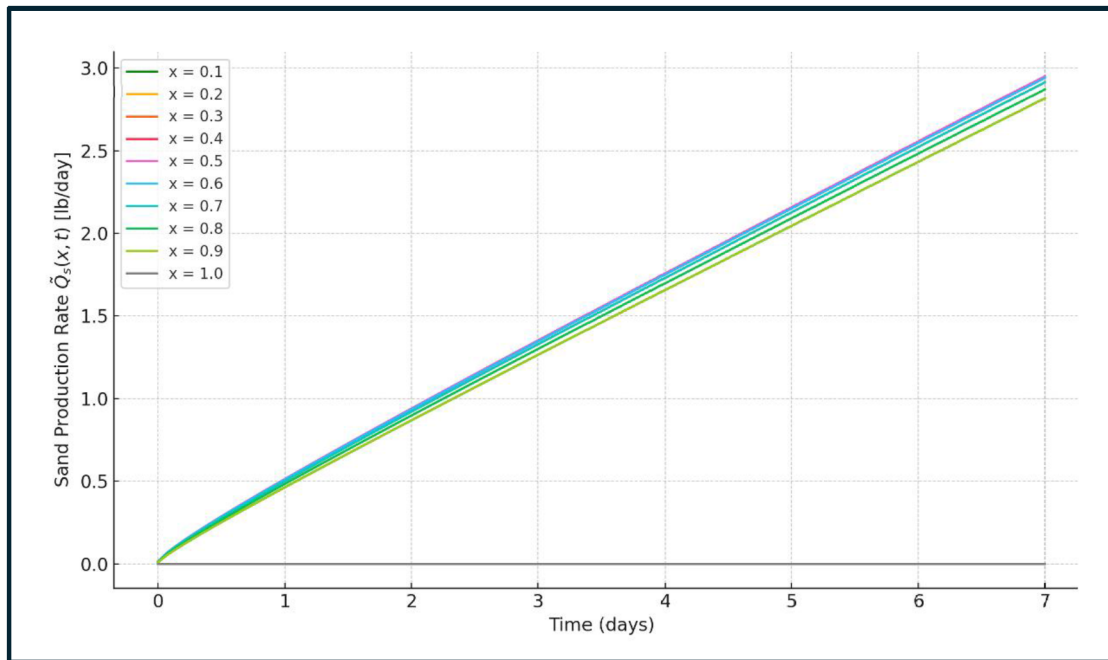


Figure 7—Sand production at different spatial locations

Fig. 7: illustrates the computed sand production rate $\dot{Q}_s(x, t)$, expressed in mass units [lb/day], over a 7-day interval across a normalized radial domain $x \in [0.1, 1.0]$. This model incorporates chemo-mechanical weakening, stress-dependent failure, and contact-driven damage evolution in weakly cemented elastic formations.

The formulation includes the effects of

- Time-evolving volumetric strain $e_v(x, t)$
- Chemical and mechanical damage terms $D_{chemo}(x, t)$ and $D_{mech}(x, t)$
- AI/ML-informed microdamage correction $\Gamma_{AI/ML}$
- Fluid-induced erosion proportional to velocity gradients ∇V_f
- A compaction-resistance term $C(x, t)$

- And critically, the spatial-temporal yielded zone $l_{yield}(x, t)$, defined from elasto-plastic FEM analysis.

The results show that sand production initiates and intensifies in regions where local stress fields exceed the yield threshold, and where damage has sufficiently progressed. Specifically:

- Mid-radial zones ($x = 0.4$ to 0.6) exhibit the highest production rates due to an optimal overlap of damage accumulation, volumetric strain, and active stress concentrations. These regions likely represent the transition zone between peak tangential stress and damage localization.
- Far-field regions ($x = 1.0$) remain un-yielded throughout the simulation, as expected, due to insufficient stress amplification and low fluid-shear gradients. This confirms that the model preserves the elastic domain.
- The near-wellbore wall ($x = 0.1$) surprisingly demonstrates lower-than-expected sanding rates. This result can be physically interpreted as the consequence of:
 1. High confining stress suppressing compressive failure,
 2. Early stress arching or redistribution,
 3. Reduced effective drag due to overbalanced pressure conditions near the wellbore.

The monotonic increase in all active x-zones confirms the progressive and irreversible nature of the sand production process under coupled chemo-mechanical loading.

Conclusion and Recommendations

Conclusion

This study comprehensively investigated the effects of drilling fluid exposure on the mechanical integrity and sanding potential of weak clastic sandstone reservoirs. Experimental results from advanced thick-wall cylinder tests (ATWC), mineralogical and chemical analyses, and poromechanical modeling demonstrated that drilling fluid exposure significantly weakens the rock strength of the rock matrix. Strength reductions ranging from 15% to 23% were observed, with the most severe degradation associated with high-viscosity and electrically stable fluids. Mineral dissolution, clay swelling, and capillary effects were identified as the primary mechanisms driving damage, leading to increased porosity, decreased permeability, and sand mobilization.

Simulation results revealed that chemo-mechanical damage accumulates rapidly during the early exposure period and becomes dominant beyond Day 4. Volumetric strain, bond weakening, and fluid erosion collectively contributed to increased sand production rate, with maximum production localized near the mid-radial zones of the wellbore. The FEM-based sand production model incorporating strain, damage, and AI/ML correction terms successfully predicted spatial and temporal variations in sanding risk.

Recommendations

Based on the experimental and simulation outcomes, the following recommendations are proposed to mitigate sanding in weak sandstone reservoirs:

1. Carefully evaluate and optimize drilling fluid formulations to minimize chemical reactivity, especially with swelling clays and feldspars.
2. Use formation-compatible fluids with lower electrical stability and reduced plastic viscosity to limit deep filtrate invasion.
3. Implement pre-exposure core screening using TWC or ATWC tests to benchmark rock strength sensitivity to drilling fluid exposure.

4. Integrate real-time geomechanical monitoring with predictive sand production modeling to adjust drawdown strategies dynamically.
5. Utilize FEM and AI-based workflows to design completions that avoid sanding-prone zones based on porosity, permeability, and cohesion maps.
6. Avoid aggressive production startup; implement managed drawdown to allow stress redistribution without exceeding sanding thresholds.
7. Consider chemical stabilizers or resin-based additives for formations with high damage indices to reinforce grain bonding before production.

References

1. Addington, D. V., Avasthi, J. M., & Schmidt, S. D. (1991). "Formation damage by drilling and completion fluids" *SPE Formation Damage Control Conference*.
2. Ahad, A. (2020). *Mechanisms of Sand Production in Weak Sandstones*. SPE Technical Paper.
3. Ahad, R. A. (2020). *Sand Production Mechanisms and Control in Oil and Gas Reservoirs*. Springer.
4. Al-Ajmi, A. M., & Zimmerman, R. W. (2005). "Relation between the Mogi and the Coulomb failure criteria". *International Journal of Rock Mechanics and Mining Sciences*, **42**(3), 431–439. <https://doi.org/10.1016/j.ijrmms.2004.11.004>
5. Al-Ajmi, A. M., & Zimmerman, R. W. (2005). Stability analysis of vertical boreholes using the Mogi–Coulomb failure criterion. *International Journal of Rock Mechanics and Mining Sciences*, **42**(3), 431–439. <https://doi.org/10.1016/j.ijrmms.2004.11.001>
6. Al-Yaseri, A., et al (2023). "On hydrogen–cement reaction: investigation on well integrity during underground hydrogen storage". *International Journal of Hydrogen Energy*, **48**(91), 35610–35623.
7. Fjær, E., Holt, R. M., Horsrud, P., Raaen, A. M., & Risnes, R. (2008). *Petroleum Related Rock Mechanics*. Elsevier.
8. Hamid, O., & Oluyemi, G. (2022). Elastic Modeling of Sand Production. *InterPore Conference*.
9. Hamid O., and Alfaraj, R. 2023. Solid Production Prediction in Weak Sandstone Formation Using Finite Element Modeling (FEM) Approaches. ARMA-IGS-2023-0130. This paper was prepared for presentation at the International Geomechanics Symposium (IGS) on 30 October - 2 November 2023 in Al Khobar, Saudi Arabia
10. Jaeger, J. C., Cook, N. G. W., & Zimmerman, R. (2007). *Fundamentals of Rock Mechanics* (4th ed.). Wiley-Blackwell.
11. Liou, J. C. P. (2014). "Sand Production Management for Unconsolidated Sandstone Reservoirs". *Journal of Petroleum Science and Engineering*, **120**, 140–149.
12. Morita, N., Black, A. D., & Fuh, G. F. (1996). "Borehole stability considerations in designing deviated wells". *SPE Drilling & Completion*, **11**(02), 109–117.
13. Papamichos, E., Malmanger, A., & Brignoli, M. (2001). "Sand production and loss of well integrity". *SPE International Symposium on Formation Damage Control*.
14. Pham, H. T. (2017). Sand Production Mechanisms and Mitigation Techniques. *JPT*.
15. Schlumberger (2009). Mechanical Earth Model Report for Well MC-519 #1-BP1.
16. Surej Kumar, K. (2020). Sand production mechanism and its impact on reservoir performance: A review. *Journal of Petroleum Science and Engineering*, **189**, 107003. <https://doi.org/10.1016/j.petrol.2020.107003>
17. Tao, W., Rahman, S. S., & Rahman, M. M. (2008). "Sand production prediction and management in weak sandstone reservoirs". *Journal of Petroleum Science and Engineering*, **60**(1), 29–39.
18. Yang, X. et al (2016). Advanced Coupled Flow–Geomechanics Simulation. *SPE Journal*.

19. Zhang, J., Fan, Q., & Ghassemi, A. (2011). "Effect of drilling fluid invasion on the strength and mechanical behavior of shales". *Journal of Petroleum Science and Engineering*, **78**(3-4), 451–460. <https://doi.org/10.1016/j.petrol.2011.06.003>
20. Zhang, J., Zhang, J., & Roegiers, J.-C. (2011). Drilling Fluid-Induced Alteration of Rock Properties and Its Impacts on Sanding. *SPE Drilling & Completion*, **26**(01), 22–31. <https://doi.org/10.2118/115475-PA>
21. Zoback, M. D. (2010). *Reservoir Geomechanics*. Cambridge University Press.